

Graphene takes a step towards renewable fuel

Researchers at Linköping University are working towards developing a method to convert water and carbon dioxide into the renewable energy of the future, using energy from the sun and graphene applied to the surface of cubic-silicon-carbide.

The research group has developed a method that makes it possible to produce graphene with several layers in a tightly-controlled process. The group has also shown that graphene acts as a superconductor in certain conditions.

Carbon, oxygen and hydrogen are the three elements one would get when molecules of carbon dioxide and water are taken apart. These elements are the building blocks of chemical substances that are used for fuel, such as ethanol and methane. Conversion of carbon dioxide and water into renewable fuel, if possible, would provide an alternative to fossil fuels, and contribute to reducing carbon-dioxide emission from the atmosphere. Jianwu Sun, senior lecturer at Linköping University, is trying to find a way to do just that.

The first step is to develop the material. Linköping University researchers previously developed a method to produce cubic-silicon-carbide, which consists of silicon and carbon. The cubic form can capture energy from the sun and create charge carriers. However, this is insufficient.

Graphene, one of the thinnest materials, plays a key role in the project. It comprises a single layer of carbon atoms bound to each other in a hexagonal lattice. It can also conduct an electric current—a property that would be useful for solar energy conversion. Graphene also has several unique properties, and possible uses of graphene are being extensively studied all over the world.

Water-based fuel to power electric vehicles

Australian-Israeli startup Electriq ~ Global has developed a fuel that is safe, cheap and clean. Comprising 60 per cent water, Electriq's revolutionary technology extracts hydrogen from the water-based fuel, which is then harnessed to create electricity to power vehicles.

Electriq ~ Global system contains three key elements. Liquid fuel (Electriq ~ Fuel) reacts with a catalyst (Electriq ~ Switch) to release hydrogen on demand. Then, spent fuel is captured and taken back to a plant, where it is replenished with hydrogen and water for re-use (Electriq ~ Recycling). This entire process is inherently safe and enables zero-emissions vehicles.

The water-based liquid fuel is safe and stable at ambient temperatures and pressures. When compared to green energy competitors like lithium-ion batteries or compressed hydrogen technologies, Electriq ~ Global achieves twice the range at half the cost. Energy density is up to 15 times that of electric batteries currently in use in vehicles like those of Tesla.

A comparison done between electric buses showed that buses powered by batteries provided a range of 250km and required up to 300 minutes to recharge. Whereas, buses powered by Electriq ~ Fuel provided a range of 1000km and could be refuelled in five minutes.

Innovation leads to smaller, cheaper and long-lasting IoT sensors

Researchers from Green IC research group at National University of Singapore (NUS) have invented a low-cost, battery-less wake-up timer in the form of an on-chip circuit. It significantly reduces power consumption of silicon chips for Internet of Things (IoT) sensor nodes. The timer demonstrates for the first time the achievement of power consumption down to true picoWatt range—one billion times lower than a smartwatch.

Team leader Massimo Alioto, who is associate professor in Department of Electrical and Computer Engineering at NUS Faculty of Engineering, said, "We have developed a novel wake-up timer that operates in picoWatt range, and cuts power consumption of rarely-active IoT sensor nodes by 1000 times.

"As an element of uniqueness, our wake-up timer does not need any additional circuitry, as opposed to conventional technologies, which require peripheral circuits consuming at least 1000 times more power (for example, voltage regulators). This is a major step towards accelerating the development of IoT infrastructure, and paves the way for aggressive miniaturisation of IoT devices for long-lasting operations."

Mobile device to identify airborne allergens using deep learning

Researchers at UCLA have invented a portable device that uses holograms and machine learning to identify and measure airborne biological particles, or bioaerosols, that originate from living organisms such as plants or fungi. The system is trained to recognise five common allergens, namely, pollen from Bermuda grass, oak, ragweed and spores from two types of mould. It classifies samples with an accuracy of 94 per cent using deep learning.



The device weighs less than 600 grams, and measures about 14 centimetres wide, 17 centimetres long and six centimetres thick (Credit: UCLA Newsroom)

The device pulls in air and traps particles on a sticky surface that is lit up by a laser, which generates a hologram. An image sensor chip scans the hologram and sends that data to a remote server. There, a type of artificial intelligence (AI) powered by a neural network cleans up the image, which is then run through an algorithm that crops it down to sections that depict biological particles. A second neural network classifies those particles from among a set of preloaded allergen types.